THE IMAGE OF A PRO-AFFINE SCHEME IN A JACOBSON GROUP SCHEME WITH RATIONAL CLOSED POINTS LANDS IN AN AFFINE OPEN

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Proposition proposition-connected-component (canonical tag 0B7R): the connected component G^0 is a closed subgroup scheme and components are translates. cited at proof lines 1318-1319. Supplies G^0 for the component-wise reduction.
- Lemma limits-lemma-topology-limit (canonical tag 01YY), Definition topology-definition-profinite (08ZX), Lemma topology-lemma-spectral-pi0 (0906) with properties-lemma-quasi-compact-quasi-separated-spectral (094L). cited at proof lines 1322-1337. T and pi_0(W) are profinite, which drives the finite clopen decomposition.
- Lemma topology-lemma-closed-map (canonical tag 08YD) and topology-lemma-profinite-refine-open-covering (08ZZ). cited at proof lines 1340-1349. Closedness of psi produces the neighbourhoods; the refinement makes the cover finite and disjoint.
- Lemma lemma-group-scheme-over-field-translate-open (canonical tag 0B7N). cited at proof lines 1327-1329. Makes W invariant under left translation by G^0.
- Lemma lemma-profinite-product-over-field (canonical tag 0B7T). cited at proof lines 1382-1386. Identifies |U x_k T| = |U| x |T| so the image of the open immersion is a finite union of boxes.

Source reference. Original proposition 0B7R: sources/stacks-groupoids/pinned-original-source.tex, source lines 1230–1230. Cited at proof lines 1318-1319 to supply the closed subgroup scheme G^0 used in the two reduction paragraphs; load-bearing.

Source reference. Original lemma 01YY: sources/stacks-chow/tags.txt, source lines 2573–2573. Cited at proof line 1322 to see T is a directed inverse limit of finite topological spaces; load-bearing for the profinite observation.

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Source reference. Original definition 08ZX: sources/stacks-chow/tags.txt, source lines 11390–11390. Cited at proof lines 1322-1324 to name the topology of T profinite; framing of the topological argument.

Source reference. Original lemma 0B7N: sources/stacks-groupoids/pinned-original-source.tex, source lines 965–965. Cited at proof lines 1327-1329 to make W invariant under left translation by G^0; load-bearing.

Source reference. Original lemma 0906: sources/stacks-chow/tags.txt, source lines 11399–11399. Cited at proof lines 1334-1337 (with 094L) to see pi_0(W) is profinite; load-bearing for the finite clopen refinement.

Source reference. Original lemma 094L: sources/stacks-chow/tags.txt, source lines 11554–11554. Cited at proof lines 1334-1337 (with 0906) to see pi_0(W) is profinite; load-bearing for the finite clopen refinement.

Source reference. Original lemma 08YD: sources/stacks-chow/tags.txt, source lines 11336–11336. Cited at proof lines 1340-1342 to see psi is closed, whence the neighbourhoods V_xi exist; load-bearing.

Source reference. Original lemma 08ZZ: sources/stacks-chow/tags.txt, source lines 11392–11392. Cited at proof lines 1346-1349 to replace the covering by V_1, ..., V_n; load-bearing.

Source reference. Original lemma 0B7T: sources/stacks-groupoids/pinned-original-source.tex, source lines 1269–1269. Cited at proof lines 1382-1386 to read |U x_k T| = |U| x |T|, whence the image open immersion is a finite union of boxes; load-bearing.

Lemma 0.2. *Let k be an algebraically closed field. Let G be a group scheme over k. Assume that G is Jacobson and that all closed points are k-rational. Let $T = \mathrm{Spec}(A)$ where A is a directed colimit of algebras which are finite products of copies of k. For any morphism $f : T \to G$ there exists an affine open $U \subset G$ containing $f(T)$.*

Proof. Let $G^0 \subset G$ be the closed subgroup scheme found in Proposition 0B7R. The first two paragraphs serve to reduce to the case $G = G^0$.

Observe that T is a directed inverse limit of finite topological spaces (Limits, Lemma 01YY), hence profinite as a topological space (Topology, Definition 08ZX). Let $W \subset G$ be a quasi-compact open containing the image of $T \to G$. After replacing W by the image of $G^0 \times W \to G \times G \to G$ we may assume that W is invariant under the action of left translation by G^0, see Lemma 0B7N. Consider the composition

$$\psi = \pi \circ f : T \xrightarrow{f} W \xrightarrow{\pi} \pi_0(W)$$

The space $\pi_0(W)$ is profinite (Topology, Lemma 0906 and Properties, Lemma 094L). Let $F_\xi \subset T$ be the fibre of $T \to \pi_0(W)$ over $\xi \in \pi_0(W)$. Suppose that for every $\xi \in \pi_0(W)$ we have found affine open $U_\xi \subset W$ such that $f(F_\xi) \subset U_\xi$. As ψ is closed by Topology, Lemma 08YD, $\psi(T \setminus f^{-1}(U_\xi))$ is a closed subset of $\pi_0(W)$ that does not contain ξ. Hence its complement V_ξ is an open neighborhood of ξ and $\psi^{-1}(V_\xi) \subseteq f^{-1}(U_\xi)$. The open covering of $\pi_0(W)$ by the V_ξ may be refined by a finite covering $\pi_0(W) = V_1 \amalg \ldots \amalg V_n$ of pairwise disjoint open and closed subsets (Topology, Lemma 08ZZ). For every $i = 1, \ldots, n$ pick $\xi_i \in \pi_0(W)$ such that $V_i \subset V_{\xi_i}$. As V_i is closed and U_{ξ_i} affine, $U_{\xi_i} \cap \pi^{-1}(V_i)$ is affine as well. Hence

replacing U_{ξ_i} by $U_{\xi_i} \cap \pi^{-1}(V_i)$ we have $\psi^{-1}(V_i) = f^{-1}(U_{\xi_i})$. As the V_i's cover $\pi_0(W)$, the $f^{-1}(U_{\xi_i})$'s cover T and so $f(T) \subseteq U = \bigcup_i U_{\xi_i}$. Then U is the disjoint union of finitely many affine opens and hence affine as desired.

Let Z be a connected component of G which meets $f(T)$. Then Z has a k-rational point z (because all residue fields of the scheme T are isomorphic to k). Hence $Z = G^0 z$. By our choice of W, we see that $Z \subset W$. The argument in the preceding paragraph reduces us to the problem of finding an affine open neighbourhood of $f(T) \cap Z$ in W. After translation by a rational point we may assume that $Z = G^0$ (details omitted). Observe that the scheme theoretic inverse image $T' = f^{-1}(G^0) \subset T$ is a closed subscheme, which has the same type. After replacing T by T' we may assume that $f(T) \subset G^0$. Choose an affine open neighbourhood $U \subset G$ of $e \in G$, so that in particular $U \cap G^0$ is nonempty. We will show there exists a $g \in G^0(k)$ such that $f(T) \subset g^{-1}U$. This will finish the proof as $g^{-1}U \subset W$ by the left G^0-invariance of W.

The arguments in the preceding two paragraphs allow us to pass to G^0 and reduce the problem to the following: Assume G is irreducible and $U \subset G$ an affine open neighbourhood of e. Show that $f(T) \subset g^{-1}U$ for some $g \in G(k)$. Consider the morphism

$$U \times_k T \longrightarrow G \times_k T, \quad (t, u) \longrightarrow (uf(t)^{-1}, t)$$

which is an open immersion (because the extension of this morphism to $G \times_k T \to G \times_k T$ is an isomorphism). By our assumption on T we see that we have $|U \times_k T| = |U| \times |T|$ and similarly for $G \times_k T$, see Lemma 0B7T. Hence the image of the displayed open immersion is a finite union of boxes $\bigcup_{i=1,\ldots,n} U_i \times V_i$ with $V_i \subset T$ and $U_i \subset G$ quasi-compact open. This means that the possible opens $Uf(t)^{-1}$, $t \in T$ are finite in number, say $Uf(t_1)^{-1}, \ldots, Uf(t_r)^{-1}$. Since G is irreducible the intersection

$$Uf(t_1)^{-1} \cap \ldots \cap Uf(t_r)^{-1}$$

is nonempty and since G is Jacobson with closed points k-rational, we can choose a k-valued point $g \in G(k)$ of this intersection. Then we see that $g \in Uf(t)^{-1}$ for all $t \in T$ which means that $f(t) \in g^{-1}U$ as desired. □